

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2306

University 230 kV
March 02, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2306. The Interconnection Customer proposes a 55.8 MW battery energy storage facility interconnecting at the University 230 kV substation in the Indiana area. The request was received on April 15, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 455 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$95,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Laureles / Big Oaks 100 KV	Dekalb / Jones Chapel 500 KV Line	114.0	396	6.0
Ward Gulf / Laureles 100 KV	Dekalb / Jones Chapel 500 KV	109.3	439	6.0
Minnehaha / Sta S Van Nuys 230 KV	Troy Energy Llc / Richland 500 KV	103.8	1,890	15.8
Big Oaks / Cypress Creek 100 KV	Jones Chapel / Dekalb 500 KV	100.6	422	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Laureles / Big Oaks 100 KV	Rebuild line with higher-capacity conductor	\$5,594,000	1% (\$42,000)
Ward Gulf / Laureles 100 KV	Reconductor line	\$1,713,000	1% (\$13,000)
Sta S Van Nuys / Mnnehaha 230 KV	Reconductor line	\$5,135,000	0% (\$27,000)
Big Oaks / Cypress Creek 100 KV	Reconductor line	\$1,780,000	1% (\$13,000)
	Total allocated to project		\$95,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.